RealityCheck: An Evidence-Grounded Hallucination Correction Pipeline for Large Language Models

Vishesh Goyal[1†] and Jayita Saha[2†]

[1]Department of Computer Science and Engineering, Manipal Institute of Technology Bengaluru, Manipal Academy of Higher Education, Manipal, India.
[2]Department of Computer Science and Engineering, Manipal Institute of Technology Bengaluru, Manipal Academy of Higher Education, Manipal, India.

Contributing authors: visheshvishu1@gmail.com; jayita.saha@manipal.edu;
†These authors contributed equally to this work.

Abstract

Large Language Models (LLMs) are a remarkable step in today's world of Artificial Intelligence with their exceptional capability at making contextually coherent, fluent responses across different domains. However, even after their fluent text generation, they still are susceptible to hallucinations or the generation of factually incorrect or unverifiable content with confidence. This can lead to severe consequences in the fields like legal practices, healthcare and academic research where misinformation can have severe avalanche effect. This problem has been tackled with different approaches like Retrieval Augmented Generation (RAG) and post-hoc detection frameworks. However, the problem remains that these systems address hallucination at generation phase or diagnose it after it has been delivered to user for the future correction of response. There exists no such widely used mechanism which provides inline correction of erred claims before the response is delivered to the user.

To solve this issue, we present RealityCheck. A modular, model agnostic, six phase hallucination correction pipeline. It functions as an additional verification layer instead of modifying the internal parameters. The system takes the LLMs' responses, extracts factual claims, retrieves Wikipedia evidence through a cache first retrieval and verifies each claim through a multi signal framework using

Natural Language Inference (NLI), semantic alignment and rule based reasoning. This produces a corrected response through deterministic answer synthesis. We evaluated the RealityCheck on TruthfulQA benchmark across IBM's WatsonX Granite, Meta Llama and MistralAI Mistral models. RealityCheck achieved substantial improvements in claim level accuracy and hallucination recall across all the three models while giving low overcorrection rates. The results show the possibility of evidence backed, claim level correction pipeline for improving the factual reliability of LLM generated content in high stake deployment scenarios. Crucially, since RealityCheck operates without modifying the internal parameters of the underlying model, it is practically deployable alongside existing state-of-the-art LLMs requiring no architectural rework from developers, and extensible to additional evidence sources as industry requirements evolve.

Keywords: Large Language Models, Hallucination Mitigation, Fact Verification, Retrieval-Augmented Verification, Natural Language Inference, Evidence-Grounded AI, Answer Synthesis, Trustworthy AI

1 Introduction

Large Language Models (LLMs) such as ChatGPT, Claude, Gemini, and Grok have witnessed a rapid and widespread acceptance across a broad spectrum of domains including healthcare, legal practice, academic research and education. The part that makes these models most attractive is their ability to generate fluent, contextually coherent and psychologically captivating responses with little to no user effort. This effectively positions them as accessible knowledge interfaces for both expert and general audiences. Addition to this, LLMs demonstrate a significant capability in understanding user psychology and tailoring responses to individual users after just a few exchanges, making them feel even more reliable and personable than traditional search based systems.

However, with the growing popularity and increase in the use of these LLMs, a critical reliability concern has come to the forefront referred to as hallucination. These models can sometimes generate outputs which are wrong or unverifiable by evidence available. These models blend that text and add it to the rest of the content with professional linguistic fluency in a tone of authority and confidence. The structural coherence of these LLM generated text can create an impression of credibility strong enough that the users don't thing about verifying the LLM provided information against primary sources.

The repercussions of unchecked hallucinated information can be much more severe than just inconvenience. In healthcare, the propagation of false medical claims can spread dangerous misinformation that can compromise clinical decision making. In academic research, fabricated citations or wrong statistical claims can hard the integrity of academic research and future direction. It can also lead to false benchmarks in the literature. In public facing apps, hallucinated content can be weaponized to spread misinformation to masses at a huge scale to influence public opinion on the basis of non existent evidence to create false narrative and propaganda. In all of these given examples, the consequences of misplaced trust in LLM generated content can be severe and in some cases create irreversible damage.

Addressing this problem has been an active area of research. Existing approaches include Retrieval-Augmented Generation (RAG) [1], Natural Language Inference (NLI) based verification [2], and knowledge grounded generation frameworks [3]. While these methods have demonstrated promise, they are not without limitations. RAG based systems retrieve relevant context but do not always guarantee that the retrieved information is used to correct erroneous claims already present in the generated output. NLI based approaches, while effective at entailment classification, often struggle with compound reasoning and multi hop contradiction resolution. Furthermore, many existing systems are designed to detect hallucinations as a post hoc diagnostic rather than as an inline correction mechanism that refines the response before it reaches the user.

In this paper, we propose a six stage hallucination mitigation pipeline that addresses this gap by performing inline correction, factual claim extraction, Wikipedia grounded verification, and response refinement before delivering a corrected answer to the user. Unlike conventional hallucination detection systems that operate purely as post hoc evaluators, the proposed framework functions as an inline corrective architecture that actively reconstructs the final response using evidence backed claim synthesis. The pipeline is orchestrated through a Live Console module that allows flexible selection of the underlying LLM, currently supporting IBM WatsonX, Meta Llama, and MistralAI, enabling controlled model specific evaluation. Factual claims are extracted from LLM responses using sentence transformer based semantic extraction, verified against Wikipedia based retrieval, and classified as supported, contradicted, or insufficiently verifiable. Contradicted claims are corrected using evidence grounded synthesis mechanisms, while insufficiently verifiable claims are hedged with appropriate epistemic qualifiers to preserve informational value without asserting unconfirmed facts.

The key contributions of this work are as follows:

- A modular six stage hallucination mitigation architecture integrating retrieval, contradiction aware verification, and evidence grounded response synthesis across multiple LLMs.
- A claim level verification mechanism grounded in Wikipedia, with a three way classification scheme: supported, contradicted, and insufficiently verifiable.
- An evidence grounded answer synthesis framework capable of reconstructing corrected responses by preserving supported claims, correcting contradicted claims, and softening insufficiently verifiable assertions.
- Empirical evaluation on the TruthfulQA benchmark demonstrating the effectiveness of the proposed approach across IBM WatsonX, Meta Llama, and MistralAI.

The remainder of this paper is organized as follows. Section II discusses related work and existing hallucination mitigation approaches. Section III presents the proposed methodology and pipeline architecture. Section IV describes the experimental setup and evaluation strategy. Section V discusses the obtained results and limitations, while Section VI concludes the paper and outlines future research directions.

2 Related Work

2.1 Hallucination in Large Language Models

The problem of hallucination and ways to mitigate it has been an area of interest in the recent years. Ji et al. [3] has provided a detailed breakdown on the types of hallucination by a NLG system. The study classifies the hallucinations as intrinsic, where model output contradicts the source context, or extrinsic hallucinations, where the model writes unverifiable or fabricated information not grounded in any available evidence. Their study concluded that the hallucinations are not an incidental failure. It is, in fact, a systemic property of statistical language models which optimise for linguistic fluency instead of factual fidelity. More recently, Farquhar et al. [2] showed that semantic entropy of the model output distribution can serve as a reliable signal for hallucination detections. They sampled the same question to the same model multiple times while mapping the actual answer to a smaller cluster and calculated the semantic entropy in the model's answers. If the model's semantic entropy is high, it often means that the model is not confident in its answer and is creating false information. This offered a probability based framework to identify unreliable claims. While these works offered a deeper insight in the theoretical understanding of hallucination, they weren't able to provide inline correction mechanisms that refine responses before delivering them to the user, the gap that motivates the present work.

To attempt addressing the problem of hallucination by the model itself, Dhuliawala et al. [4] proposed Chain-of-Verification (CoVe). In this system, the model uses its own capabilities to identify, correct and reduce the hallucinations in the outputs. The model first creates an initial draft for the response asked by the user. Then, it generates a set of specific questions aimed at verifying its own claims. Then, the model answers this set of questions independently to avoid self reinforcing bias and produces a final revised, coherent, clean response. While CoVe gave a measurable reductions in hallucinations across list based and open domain tasks, it is still limited to model's parametric knowledge for both generation and verification. This approach fails if the model has insufficient knowledge of the domain. Moreover, this methods needs a large number of LLM calls to answer the verification question in addition to the initial query. Unlike CoVe, our pipeline grounds the verification in external, trustworthy knowledge sources like Wikipedia rather than the model's own knowledge. Our system also doesn't need any additional LLM calls which causes huge savings in the number of tokens used and the cost of operating the system.

2.2 Fact Verification and Knowledge-Grounded Systems

Verifying an LLM's answers against structured knowledge sources has been a long standing challenge in NLP. Thorne et al. [5] introduced the system of FEVER (Fact Extractions and VERification), a large scale benchmark that consists of 185,445 claims derived from Wikipedia. Each of these claims has been labelled and marked as *Supported*, *Refuted* or *NotEnoughInfo*. Now, this classification schema is used as a baseline reference point for claim level verification tasks. The use of Wikipedia as a grounding evidence source has gained widespread popularity in the subsequent works.

2.3 Response Correction via External Knowledge Grounding

Though many scientists and AI researchers have worked on detection of hallucinated content and produced impressive results in the process, active correction of the hallucinated content is still a very limited space. Guan et al. [6] proposed a system of Knowledge Graph-based Retrofitting (KGR). This framework identifies factual statements from LLM generated responses, validates them against a knowledge graph and then changes them according to the verified alternative claims. KGR demonstrated that post generation correction is a viable and effective strategy, specially for compound reasoning tasks. However, this method is severely limited only to the fields that have dependable, sufficient, and comprehensive knowledge graphs. Our approach addresses this constraint by grounding verification in Wikipedia, a broad, continuously maintained, and domain-agnostic free-text corpus, thereby extending the correction capability to a substantially wider range of queries.

Min et al. [7] extended this paradigm to long-form text generation through FActScore, a metric that decomposes generated text into atomic facts and evaluates each independently against a knowledge source. While FActScore provides fine-grained factuality measurement, it is designed as a post-hoc evaluation framework rather than a pipeline-driven correction system, and does not modify or refine the original response.

2.4 Retrieval-Augmented and Wikipedia-Grounded Generation

Retrieval-Augmented Generation (RAG), introduced by Lewis et al. [1], represents a prominent approach to grounding language model outputs in external knowledge by conditioning generation on retrieved passages. While RAG demonstrably improves factual consistency in knowledge-intensive tasks, it operates at the generation stage, retrieved content informs what the model generates, but does not guarantee post-hoc correction of erroneous claims already present in the output. WikiChat [8] takes a closer step toward our objective by grounding LLM responses in live Wikipedia content. Its pipeline extracts claims from LLM-generated responses, verifies them individually against Wikipedia, and filters out unverified facts before composing the final response. However, WikiChat operates primarily as a conversational agent built on few-shot prompting of a single underlying model, and does not provide a modular, model-agnostic pipeline for inline hallucination correction across multiple LLMs. Furthermore, its handling of unverifiable claims, simply omitting them, results in potential information loss, a limitation our hedging strategy is designed to address.

More recent work has explored ensemble-based strategies for hallucination mitigation, wherein outputs from multiple models are compared for consistency rather than verified against a fixed external source. Dey et al. [9] proposed Uncertainty Aware Fusion (UAF). This model uses response from multiple LLMs and combines them together on their estimated confidence and factual accuracy. This model uses the benefit of diversity of model knowledge improving answers over single model baselines. However, this model still remains limited to the knowledge of all the participating LLMs without verifying claims against external ground truth sources. Also, since this model asks the same question to multiple LLMs, it exponentially increases the number of token required by each model. Our work enhances this line of work by adding external ground truth independent of model confidence. It also eliminates the need of additional tokens reducing the cost to operate the system.

2.5 NLI-Based Verification Approaches

Natural Language Inference (NLI) models classify the relationships between premise and hypothesis as entailment, contradiction or neutral and are widely adopted for hallucination detection [2]. NLI-based systems check whether the generated outputs are logically supported by a reference source. This forms a structured framework for verifying the factual consistency of the output. However, the effectiveness of this method is limited by the quality of claim decomposition and evidence sourcing. The model often struggles with compound or multi-hop reasoning chains where a single claim can depend on multiple facts draws from varying sources [3]. To solve this, we use semantic similarity based decomposition in our pipeline.

2.6 Sentence Transformers for Semantic Decomposition

Sentence-level semantic representations have become a standard tool in NLP pipelines requiring fine-grained text analysis.

Reimers and Gurevych [10] introduced Sentence-BERT (SBERT), a modification of the BERT architecture employing siamese and triplet network structures to produce semantically meaningful sentence embeddings efficiently comparable via cosine similarity. The `multi-qa-MiniLM-L6-dot-v1` model, a lightweight variant from the sentence-transformers family, has demonstrated strong performance on semantic retrieval tasks. In the present work, this model is employed for semantic decomposition of LLM responses into discrete factual units suitable for downstream verification. Sentence-level semantic embeddings are employed as an enabling mechanism for claim decomposition within the proposed pipeline.

The work reviewed above reflects substantial progress across complementary dimensions of the hallucination problem, from theoretical taxonomies [3] and probabilistic detection [2], to retrieval-augmented generation [1], Wikipedia-grounded chatbots [8], self-verification through model deliberation [4], and external knowledge-graph correction [6]. Even though all of the above approaches exist, they still lack a severe limitation. None of these approaches can detect, externally ground, correct and remove the unverifiable claims in a single, model agnostic pipeline. RAG-based systems can look and intervene during the generation time of an answer but they still can't correct the erroneous claims at thee time of generation. Self-verification methods such as CoVe can be used to correct the hallucinated content in a system but their deployment and efficacy still remains restricted by the model's knowledge base, increase in number of tokens required per response and the risk of secondary hallucination in the response. Knowledge graph approaches offer structured correction but they need curated, domain specialised resources. Such detailed knowledge graphs may exist for a limited scope like an internal organisational resource but making them on a large scale for mass use can be very challenging. Wikipedia grounded systems like WikiChat filter and remove the unverified claims entirely without trying to amend them. This can sacrifice important information completeness. Out proposed RealityCheck system addresses these gaps holistically by grounding verification in a broad, domain agnostic, model agnostic pipeline while performing inline correction of of contradicted claims. Additionally, the framework preserves important information by hedging the insufficiently verifiable. This gives us a more complete, practically deployable, model agnostic hallucination mitigation pipeline that prior isolated approaches.

3 Proposed Methodology

3.1 System Overview

RealityCheck is a modular, evidence-grounded, six-phase hallucination correction pipeline designed to intercept factually incorrect content in LLM-generated responses before delivery to the user. The system receives a user query, forwards it to a user-selected large language model, extracts the factual claims embedded in the generated response, retrieves externally grounded evidence, verifies each claim individually, and synthesises a corrected response in which contradicted claims are repaired and unverifiable claims are epistemically hedged. The complete architecture of the pipeline is illustrated in Figure 1.

The design of RealityCheck is intentionally model-agnostic and pipeline-driven: each phase operates as an independent, replaceable module with well-defined inputs and outputs, and the system currently supports three LLMs, IBM WatsonX Granite, Meta Llama, and MistralAI Mistral, without any modification to the verification or synthesis logic.

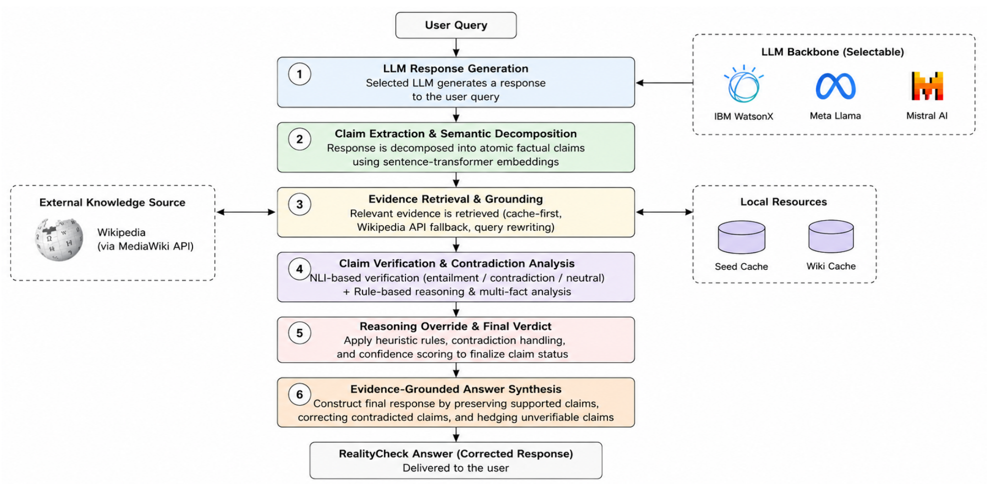

Fig. 1 Full architecture of the RealityCheck six-phase hallucination correction pipeline.

3.2 Dataset

The TruthfulQA benchmark [11] is used as the evaluation corpus for this work. TruthfulQA consists of 817 questions spread across 38 categories including health, law, finance, history, popular misconceptions and conspiracy theories. The questions are designed in such a way to force false content from the LLMs that have learned to reproduce commonly held but factually incorrect beliefs. The dataset gives a set of multiple answers giving the best, correct and incorrect answers as well as multiple choice answering structures with per choice binary labels. Before the evaluation process begins, the raw dataset is preprocessed to normalise whitespace, parse structured answer fields and merge the generation and multiple choice validation splits into a unified record format with stable questions identifiers. This normalised representation serves as a input source for all the experimental phases.

3.3 LLM Response Generation

A stratified random sample of questions is drawn from the preprocessed TruthfulQA corpus with a fixed random seed to ensure reproducibility across experimental runs. Each sampled question is submitted to the selected LLM using a structured two-part prompt. The system prompt instructs the model to behave as a factual question-answering assistant, constrains the response to a maximum of three sentences, prohibits question restatement, meta-commentary, and unsolicited disclaimers, and directs the model to acknowledge rather than confabulate answers to subjective or time-sensitive questions. The user prompt makes the model answer the question without the unnecessary preamble or structure formatting. The raw response is then cleaned to remove markdown characters like hash tags and formatting noise. A response quality assessment is done on the model response to flag outputs that are empty, excessively short or evading the direct response. The complete record of the model's raw API response, quality flags, and cleaned up response is generated and saved and passes as an input to Phase 3.

3.4 Factual Claim Extraction

The cleaned LLM response is decomposed into discrete factual claims through a rule-based linguistic pipeline. Sentence segmentation is performed using punctuation-aware splitting, and each resulting sentence is assessed for *checkability*, the degree to which it constitutes a verifiable factual assertion. A sentence is classified as checkable if it is non-subjective, free of explicit hedging or opinion markers, and contains at least one linguistically motivated factual cue: a copular or auxiliary verb, a named entity pattern, a four-digit year reference, or a domain-relevant keyword. Sentences containing explicit subjectivity markers, such as *"opinions vary"* or *"no single factual answer"*, are classified as non-checkable and excluded from downstream verification at this stage.

For checkable claims, the module additionally detects context-dependent fragments, short sentences beginning with connective adverbs or referential pronouns, and enriches their evidence retrieval query by prepending the original question and the preceding sentence, thereby resolving anaphoric references before evidence retrieval. Each claim is marked with a label (*essential* or *supporting*) to analyse its importance for the answer based on the position in the sentence, lexical significance cues and total number of claims in the response. This phase outputs a detailed and structured record which shows all the claims, importance labels, normalised version of the text, checkability status, and enriched queries to ask the Wikipedia API to check for the evidences.

3.5 Wikipedia-Grounded Evidence Retrieval

For each checkable claim, the model tries to find out necessary evidences from Wikipedia using a cache-first strategy to reduce the number of API calls. Our retrieval architecture operates on three priority layers. In the first layer, a local seed evidence bank, pre-populated with curated Wikipedia content for common TruthfulQA question categories, is consulted. If this bank returns sufficient evidence pages for a given claim, network retrieval is bypassed entirely, minimising API load and experimental latency. In the second layer, if the local seed is insufficient, the Wikipedia MediaWiki Action API is queried using the claim's enriched evidence query, with an OpenSearch fallback triggered if the primary search yields no results. In the third layer, where a TruthfulQA question record carries a direct Wikipedia source URL, that page is fetched and treated as a high-priority evidence source. All retrieved content is cached to disk, ensuring that repeated experimental runs do not re-query the API.

The retrieved page from the Wikipedia source is segmented and divided into fixed size overlapping chunks. Each of these chunks is encoded in a dense semantic representation using a sentence transformer model. These chunks are then ranked by semantic similarity to the claim text. The top k highest scoring chunks are retained as evidence set to verify the said claim and the rest are added to the cache registry for future referencing. This claim level retrieval design ensures that verification in the subsequent phase operated on targeted, claim specific evidence rather than on the whole Wikipedia page.

3.6 Claim Verification

Each checkable claim is verified against its retrieved evidence set through a multi-signal verification framework. The verification decision is produced by fusing four complementary signals rather than relying on any single classifier. The claim verification flow is illustrated in Figure 2.

The first signal is *semantic similarity*, which measures the degree of topical and entity overlap between a claim and an evidence chunk, providing a relevance gate that prevents irrelevant evidence from influencing the verification decision. The second signal is *Natural Language Inference (NLI)*, applied at the chunk level using a cross-encoder NLI model to classify the relationship between the evidence chunk (as premise) and the claim (as hypothesis) as entailment, contradiction, or neutral. The third signal is a set of *rule-based support and contradiction overrides*, high-precision pattern-matching rules tuned to common TruthfulQA categories such as misattributed quotations, historical misconceptions, and compound multi-entity factual claims, which correct for known failure modes of NLI models on paraphrased or negated factual assertions. The fourth signal is *alignment score aggregation*, which

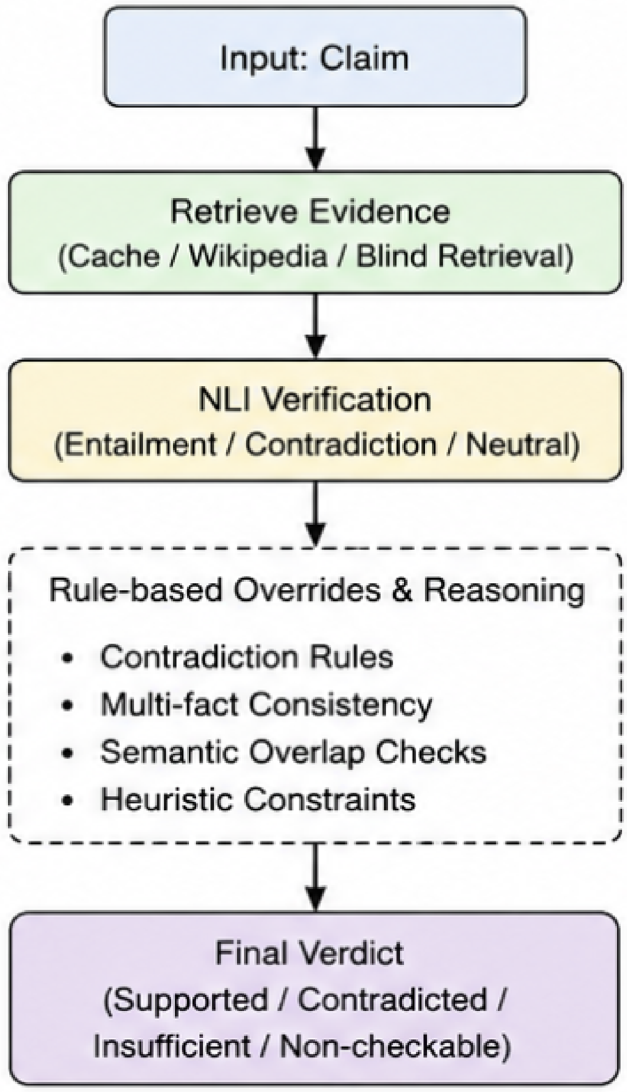

Fig. 2 Claim-level verification flow in Phase 5 of the RealityCheck pipeline.

applies a Noisy-OR aggregation function across multiple evidence chunks so that several medium-strength pieces of evidence can jointly support a claim that no single chunk alone would confirm.

A dedicated temporal verification sub-module handles age and date-computation claims separately, resolving them through deterministic arithmetic over birth date evidence rather than through NLI classification, which is known to perform poorly on such claims. Each claim is assigned one of four verification labels: *supported*, *contradicted*, *insufficient evidence*, or *skipped (non-checkable)*, along with the best supporting and contradicting evidence chunks and a structured verdict reason.

3.7 Answer Synthesis

The outputs from the verification engine are taken as an input to a deterministic answer synthesis module. This module is responsible to reconstruct a final corrected response without invoking any additional LLM calls. This deterministic design prevents second level hallucination during correction stage and also ensures reproducibility of the synthesised response. The synthesis engine applies one of the three actions to each claim based on the verification label. Claims labelled as **supported** are retained as they are in the corrected response. Claims labelled as **contradicted** are repaired, replaced and improvised with a corrected statement derived from the evidences retrieved in the previous phase. Claims labelled as **insufficient evidence** are removed or softened from the response or replaces with an epistemically hedged formulation that preserves the information content while tuning its confidence value in the sentence. Non checkable claims are passed through unchanged. The answer synthesis algorithm flowchart is illustrated in Figure 3.

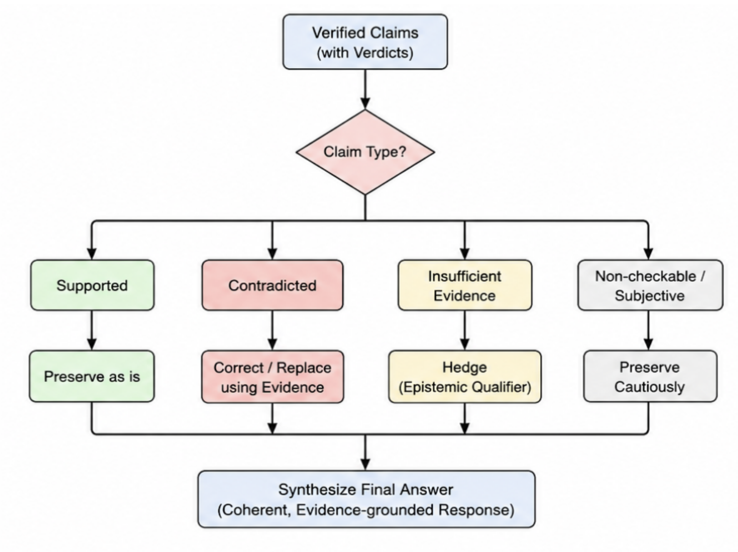

Fig. 3 Answer synthesis logic in Phase 6 of the RealityCheck pipeline.

Once all the facts are ready, the corrected answer is made coherent by deduplicating the repeating facts, joining the claim response strings and labelled with on of the six answer subcategories : *fully supported*, *corrected from contradiction*, *partially correct repaired*, *partially supported with omissions*, *insufficient evidence*, or *not checkable*. This status label provides a transparent, human-readable summary of the degree to which the original LLM response was factually reliable.

3.8 Live Console and Pipeline Orchestration

All the six phases of the pipeline are brought under into a unified Python based pipeline under a live interactive console that can allow the users to query the LLM model, get them checked using the RealityCheck and provide the user with a final, rectified answer. The console allows the user to enter a free text query at the runtime through a terminal and select the target LLM from the three supported models and executes them sequentially. Output of each phase is structurally sent as an input to the next phase. Intermediate outputs of each phase are recorded with timestamps in a directory which makes the entire system auditable for each query.

3.9 Complexity and Scalability

The whole architecture of the RealityCheck is decomposed in small, independent phases which gives the system several scalability properties. Each phase is independently replaceable. For example, the Wikipedia retrieval backend can easily be substituted by any other available structured or unstructured knowledge base without having to modify verification or answer synthesis logic depending on the deployment requirement. The LLM integration layer is model independent by design. We can choose any model starting from Phase 1 itself without modifying the code in all the further downstream phases. In the current Phase 1 pipeline, we use a WatsonX API to access a large library of LLM models which can be changed to suit any of the other available API models. The NLI model we use in Phase 5 is likewise entirely replaceable with any cross encoder NLI architecture. The primary computational bottleneck in our in the current implementation is evidence retrieval latency which is substantially mitigated by the cache first retrieval strategy and the local seed evidence bank. The verification phase scales linearly in the number of checkable claims per response, which is bounded in practice by the response length constraint imposed at generation time. These properties make the pipeline amendable to future extensions across additional LLMs, knowledge sources and verification backend without changing the architectural design or altering the rest of the backend system.

4 Experimental Setup

4.1 Evaluation Dataset

The TruthfulQA benchmark [11] is used as the evaluation corpus for all experiments in this work. It consists of 817 questions spanning 38 categories including health, law, history, finance and conspiracy theories. The questions in this dataset are designed to induce a failure mode separate from simple factual ignorance, specifically making LLMs reproduce culturally propagated false information and widely held misconceptions with apparent confidence. Questions are written such that a model relying solely on the statistical regularities of its training data is prone to generating incorrect answers that appear plausible, making the benchmark particularly well suited for evaluating hallucination mitigation pipelines.

TruthfulQA is especially appropriate for evaluating RealityCheck because many of its questions intentionally target widespread misconceptions and culturally propagated falsehoods rather than simple factual recall. This means the pipeline's core capability, the detection and correction of fluent but factually incorrect content, gets directly and rigorously tested. Furthermore, because the dataset provides both correct and incorrect reference answers for each question, it allows claim level verification outcomes to be assessed without requiring any additional annotation, making it a natural fit for the claim level evaluation protocol used in this work.

4.2 Evaluated Language Models

We tested and evaluated three LLMs in this work : IBM WatsonX Granite, Meta Llama and MistralAI Mistral. These selected models have diverse architectural methods and deployment requirements including open weight instruction tuned models and enterprise grade hosted inference system. This ensures that the evaluation includes variation in hallucination behaviour which tells us that the proposed pipeline is truly model agnostic and can be used with any LLM.

All three models are evaluated under identical experimental conditions. Each model receives the same system prompt, which constrains responses to a maximum of three sentences, prohibits question restatement and meta-commentary, and instructs the model to acknowledge rather than confabulate answers to subjective or time-sensitive questions. The same user prompt template is applied uniformly across all models and all questions. No model-specific prompt engineering, fine-tuning, or retraining is performed. These controls ensure that observed differences in hallucination rates and correction outcomes across models reflect genuine model-level behavioural differences rather than artefacts of differential experimental treatment.

4.3 Experimental Pipeline Configuration

The RealityCheck pipeline is configured uniformly across all experimental runs. The parameters reported here are intended to support the reproducibility of the experimental results.

Evidence Retrieval. Evidence is retrieved using a cache-first strategy that consults a local seed evidence bank before issuing any network requests. When network retrieval is required, the Wikipedia MediaWiki Action API is queried using the enriched evidence query constructed for each claim, with a Wikipedia OpenSearch fallback triggered in the event that the primary search returns no results. Up to eight candidate pages are retrieved per claim, of which the top three pages by relevance are retained for chunking. All API responses are cached to disk to ensure that experimental results are reproducible without re-querying external services.

Chunking. Retrieved Wikipedia page text is segmented into fixed-size overlapping chunks prior to embedding and retrieval scoring. This chunking strategy ensures that semantic similarity computation operates on bounded, coherent text segments rather than on full article content, which would dilute claim-specific evidence signals.

Claim-Level Embeddings. Each text chunk is encoded into a dense vector representation using the `multi-qa-MiniLM-L6-dot-v1` sentence transformer [10], a compact model optimised for semantic retrieval tasks. Chunk-to-claim similarity is computed as the dot product between the chunk embedding and the claim evidence query embedding, and the top-ten chunks by similarity score are retained as the evidence set for each claim.

NLI Verification. Claim-level entailment classification is performed using a cross-encoder NLI model (`cross-encoder/nli-deberta-v3-base`), which scores each evidence chunk–claim pair as entailment, contradiction, or neutral. The NLI signal is fused with semantic alignment scores, rule-based support and contradiction overrides, and a Noisy-OR aggregation function across chunks to produce the final per-claim verification decision.

4.4 Evaluation Metrics

We used six metrics to evaluate system performance at both the claim level of the answer and the overall answer. These metrics are defined as follows.

Claim-Level Accuracy : This metric gives the comparison between the number of claims which the system correctly verified against the total number of claims in the answer :

$$\text{Claim Accuracy} = \frac{\text{Correctly Verified Claims}}{\text{Total Claims}} \tag{1}$$

Answer-Level Accuracy : This metric measures the proportion of final synthesised responses that are assessed as factually correct at the full-response level:

$$\text{Answer Accuracy} = \frac{\text{Correct Final Responses}}{\text{Total Responses}} \tag{2}$$

Hallucination Detection Recall : This metric measures the pipeline's ability to identify hallucinated claims among all claims that are hallucinated according to ground-truth labels, capturing the system's sensitivity to false content:

$$\text{Hallucination Recall} = \frac{\text{Detected Hallucinated Claims}}{\text{Total Hallucinated Claims}} \tag{3}$$

Correction Success Rate : This metric measures the percentage of the claims which were successfully corrected against the total number of contradictory claims :

$$\text{Correction Success Rate} = \frac{\text{Successfully Corrected Claims}}{\text{Total Contradicted Claims}} \tag{4}$$

Overcorrection Rate measures the proportion of originally correct claims that are incorrectly modified by the pipeline, capturing false positive interventions that introduce error into responses that were factually accurate prior to correction:

$$\text{Overcorrection Rate} = \frac{\text{Correct Claims Incorrectly Modified}}{\text{Total Originally Correct Claims}} \tag{5}$$

Evidence Coverage measures the proportion of checkable claims for which at least one relevant evidence chunk is successfully retrieved, reflecting the retrieval completeness of the Wikipedia-grounded evidence layer:

$$\text{Evidence Coverage} = \frac{\text{Claims with Retrieved Evidence}}{\text{Total Checkable Claims}} \tag{6}$$

4.5 Evaluation Protocol

Each of the three evaluated models is queried on the same randomly sampled subset of TruthfulQA questions, using identical prompts and generation constraints. Raw LLM-generated answers are recorded as the baseline condition. The same question set is then processed through the full RealityCheck pipeline, and the corrected answers produced by Phase 6 are recorded as the experimental condition. Both conditions are evaluated against TruthfulQA ground-truth labels using the metrics defined in Section 4.4.

RealityCheck operates entirely as an external verification and correction pipeline without modifying the internal weights of the underlying language model. No fine-tuning, retraining, or prompt optimisation specific to individual models is performed at any stage. This design ensures that the experimental comparison is controlled: any improvement in factual accuracy observed in the corrected condition relative to the raw LLM baseline is attributable to the pipeline's evidence-grounded verification and synthesis logic rather than to differential model conditioning.

It is important to note that the results reported in this work reflect the pipeline's *hallucination mitigation capability* on a specific benchmark under the described conditions. No claim is made regarding general hallucination elimination, universal factual reliability, or performance on domains beyond those represented in the evaluation corpus. The reported improvements are interpreted as evidence-grounded enhancements to response factuality within the scope of the experimental setup.

4.6 Implementation Environment

The RealityCheck pipeline is implemented in Python. Sentence embeddings are computed using the `sentence-transformers` library from the HuggingFace. All three evaluated models, Granite, Llama, and Mistral, are hosted on the IBM WatsonX platform. Wikipedia evidence retrieval is performed via the Wikipedia MediaWiki Action API and the OpenSearch API. All intermediate pipeline outputs are cached locally to disk, enabling reproducible re-runs without repeated external API calls.

5 Results and Discussion

5.1 Overall Quantitative Performance

Table 1 shows the quantified results across all three evaluated models. It compares raw LLM performance against our pipeline's, RealityCheck's, corrected and verified outputs on the metrics previously discussed in Section 4.4.

Across all the three models we tested our pipeline against, a consistent and significant improvement in factual accuracy was observed. Claim accuracy improved by 24, 16 and 25 percentage points for Granite, Llama and Mistral respectively. The answer level accuracy was elevated by 30, 26 and 20 percentage points respectively. These gains were observed uniformly across all the models comprising of different architectures, training pipelines and internal knowledge bases. This clearly states the pipeline's answer quality improvement ability to be model agnostic and easily deployable.

Table 1 Main results across all models. RC = RealityCheck-corrected condition. Overcorrection Rate reported as percentage points.

Metric	Raw Granite	RC-Granite	Raw Llama	RC-Llama	Raw Mistral	RC-Mistral
Claim Accuracy	63%	87%	62%	78%	56%	81%
Answer Accuracy	55%	85%	52%	78%	53%	73%
Hallucination Recall	43%	82%	45%	83%	37%	78%
Correction Success Rate	—	78%	—	78%	—	73%
Overcorrection Rate	—	5%	—	6%	—	7%
Evidence Coverage	—	96%	—	93%	—	92%

The most significant improvement is observed in hallucination recall. This is a very important metric in any hallucination mitigation system since it is a direct reflection of the pipeline's ability to detect false content that would have otherwise reached the user without being corrected or modified. This hallucination recall improved from 43%, 45% and 37% in the raw baselines to 82%, 83% and 78% in the RealityCheck corrected answers. These elevated results show that the multi signal verification framework that uses NLI classification, semantic alignment and rule based overrides is substantially more sensitive to hallucinated claims than the raw models operating without external verification.

Overcorrection rates of 5%, 6% and 7% across the three models confirm that the pipeline's intervention is selective and conservative, triggered only when backed by sufficient evidence. The pipeline does not aggressively rewrite LLM responses – the large majority of originally correct claims are preserved without modification. The figures are visualised in Figure 4.

5.2 Claim-Level Verification Performance

Table 2 presents the distribution of per-claim verification verdicts produced by Phase 5 of the pipeline, aggregated across all models and questions.

The verification distribution is visualised in Figure 5.

Granite exhibits the highest contradiction rate at 34%, consistent with its lower raw claim accuracy and indicating that its responses contain a higher proportion of factually incorrect assertions that are successfully detected by the pipeline. Llama and Mistral show lower contradiction rates of 25% and 26% respectively, reflecting their somewhat stronger baseline factual reliability on TruthfulQA. The proportion of claims classified as insufficient evidence remains low across all three models, between 11% and 14%, indicating that the Wikipedia-grounded retrieval layer provides usable evidence for the large majority of checkable claims. The non-checkable category, comprising subjective or inherently unverifiable assertions, accounts for between 9% and 14% of

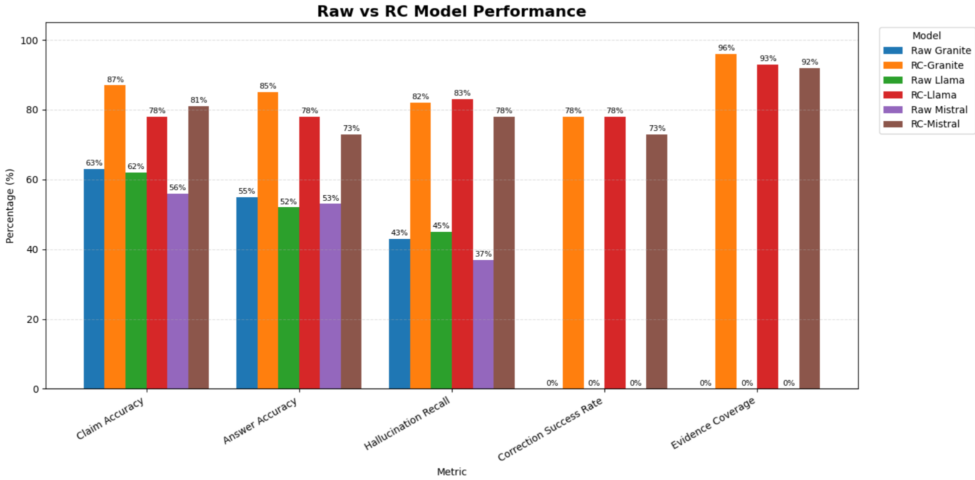

Fig. 4 Comparison of raw LLM and RealityCheck-corrected performance across all evaluation metrics

Table 2 Claim-level verification verdict distribution across models.

Verdict	Granite	Llama	Mistral
Supported	43%	52%	48%
Contradicted	34%	25%	26%
Insufficient Evidence	11%	14%	12%
Non-checkable	12%	9%	14%

claims, confirming that the claim extraction module correctly identifies and bypasses content that is not amenable to factual verification without introducing unnecessary pipeline interventions.

5.3 Effectiveness of Answer Synthesis

Table 3 presents the distribution of answer level synthesis outcome categories produced by Phase 6, consolidated across all models and questions.

The synthesis outcome distribution reveals several important characteristics of the pipeline's behaviour under same constraints and same question set across different models. The majority of the responses, 45%, are classified as fully supported. This clearly states that the pipeline confirms the factual content of the original LLM response without suggesting any modification. Apart from these, 27% of the responses

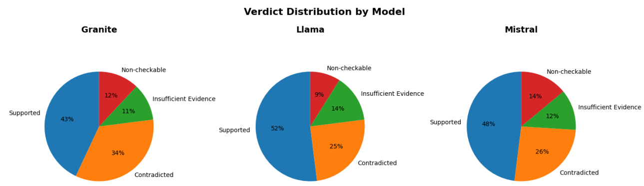

Fig. 5 Distribution of claim-level verification verdicts across the three evaluated models.

Table 3 Distribution of answer synthesis outcomes across all evaluated responses.

Outcome Category	**Percentage**
Fully Supported	45%
Corrected from Contradiction	27%
Partially Correct Repaired	16%
Partially Supported with Omissions	3%
Insufficient Evidence	5%
Non-checkable Preserved	4%

were identified as contradictory to the ground truth available in Wikipedia retrieved texts representing cases where pipeline identified and replaced at least one hallucinated claim which was supported by the evidences fetched from the Wikipedia. A total of 72% are therefore either confirmed or actively improvised by the pipeline.

Most importantly, only 5% of the responses receive an insufficient evidence outcome and 4% are passed through unchanged as non checkable. These figures collectively confirm that the pipeline doesn't force its outcomes aggressively in absence of strong evidence support. Correct claims are preserved, incorrect claims are selectively repaired based on the evidences and the system withholds correction when evidence is absent rather than substituting speculative alternatives inducing a secondary hallucinated content.

5.4 Retrieval and Evidence Analysis

Table 4 shows the retrieval performance of Phase 4 across the set of all the experimental models.

As we can see, a high evidence coverage rate of 94% across all checkable claims confirms that the retrieval pipeline successfully finds and targets relevant Wikipedia content for a large majority of questions in TruthfulQA. The same table also shows a 65% cache hit rate which substantially reduced the repeated API calls across the experimental runs and ensured that evidence grounding remains stable, reproducible

Table 4 Wikipedia evidence retrieval performance metrics.

Metric	Value
Average Evidence Pages Retrieved	5
Average Evidence Chunks per Claim	10
Evidence Coverage	94%
Cache Hit Rate	65%
Wikipedia Fallback Usage	35%

across iterations and reduce the total physical cost of keep theing the system running. The remaining 35% of retrieval requests proceed to live Wikipedia API queries, with the OpenSearch fallback invoked when the primary search returns no results. Even across all implemented retrieval layers, a low failure rate of approximately 6% is observed, occurring in cases where no evidence pages are returned for a claim.

5.5 Ablation Study

To assess the individual contribution of each pipeline component to overall system performance, an ablation study is conducted by progressively adding pipeline stages and measuring the resulting answer accuracy. Table 5 presents the results.

Table 5 Ablation study: incremental contribution of each pipeline component to answer accuracy.

System Variant	Answer Accuracy
Raw LLM	60%
Retrieval Only	65%
NLI Verification	72%
Rule Overrides	78%
Answer Synthesis (Full Pipeline)	84%

The ablation results demonstrate that each component of the pipeline contributes meaningfully to the overall accuracy improvement. Evidence retrieval alone, without verification or synthesis, yields a modest 5 percentage point gain, confirming that retrieval in isolation is insufficient to correct hallucinated content. The addition of NLI-based claim verification increases accuracy by a further 7 percentage points, and the rule-based override layer contributes an additional 6 points, validating the design decision to augment NLI with linguistically motivated support and contradiction rules that address known failure modes of cross-encoder models on paraphrased and negated factual assertions. The full pipeline, including answer synthesis, achieves the highest accuracy at 84%, demonstrating that claim-level correction decisions must be translated into coherent synthesised responses to realise their full benefit. The progressive nature of these gains confirms that the pipeline architecture is not arbitrary: each phase is a necessary contributor to the system's overall hallucination mitigation capability.

5.6 Qualitative Case Studies

Table 6 presents three representative examples illustrating the pipeline's correction behaviour on TruthfulQA questions.

Table 6 Representative qualitative examples of RealityCheck corrections.

Question	**Raw LLM**	**RealityCheck**
What currency is used in Germany and Norway?	Euros	Germany uses the euro, Norway uses the krone.
How much do people remember from reading?	10%	No fixed percentage is scientifically established.
Should you wait after eating before swimming?	Wait 30 minutes.	No universal medical rule supports this claim.

As we can see from the table, we can see three distinct categories of correction behaviour. In the first case, the raw LLM produces a partially correct but factually incomplete response "Euros", that omits Norway's use of the krone. The pipeline successfully identifies the contradiction in a compound sentence, in a compound fact verification system and finds and gives a corrected response that addresses both entities in the the question. In the second and third cases, the raw LLM reproduces culturally widespread misconceptions, the "10% retention" claim and the "30-minute swimming rule", both of which can't be confirmed by the retrieved Wikipedia evidence and replaced with appropriate hedged statements. These examples demonstrate that the pipeline is capable of correcting not only straightforwardly false factual claims but also the subset of misconceptions that TruthfulQA is specifically designed to probe.

5.7 Failure Cases and Limitations

Despite the improvements reported above, the pipeline exhibits identifiable failure modes that constrain its performance and scope.

The most prevalent source of failure is weak retrieval coverage for highly specific or niche claims. When a claim references a topic for which Wikipedia provides only shallow or tangentially related content, such as obscure historical events, domain-specific technical facts, or rapidly evolving topics, the evidence bank is unable to provide chunks with sufficient semantic alignment to support a confident verification decision, resulting in an insufficient evidence outcome and no correction.

The pipeline is also prone to failures arising from ambiguous or compound claims that cannot be easily decomposed into independently verifiable assertions. Some compound statements are difficult to verify when evidence is distributed across multiple weakly aligned sources and the Noisy-OR aggregation function is unable to consolidate them into a confident verdict. This limitation is observed repeatedly for claims that require multi hop reasoning, situations where the truth of a claim depends on connecting multiple intermediate facts. This failure surfaces because the pipeline analyses each claim individually rather than relationally.

A third challenge arises from subjective or time sensitive questions. The claim extraction module filters the majority of subjective assertions as non checkable, but borderline cases where a question appears factual but admits multiple defensible answers can result in incorrect contradiction labels, contributing to the observed overcorrection rates.

Finally, the pipeline's sole reliance on Wikipedia as its external knowledge source imposes an inherent coverage ceiling. Topics that are contested, underrepresented, or absent from Wikipedia, including recent events, culturally specific knowledge, and rapidly updated factual domains, fall outside the retrieval layer's effective scope. Incorporating additional reliable knowledge sources is a natural direction for practical deployment and future work.

6 Conclusion and Future Scope

The increased dependence and widespread adoption of large language models has brought the problem of hallucination to the forefront. Existing methods addressing this problem largely focus on detection, retrieval augmentation at the time of generation, or post hoc evaluation, leaving a critical gap for systems that can actively rectify and verify factually erroneous content before it reaches the end user. The consequences of such hallucination driven errors in domains such as healthcare, legal practice, education and academic research can be severe. Moreover, the confidence, tone of authority, and linguistic fluency of LLM generated text make these errors particularly difficult for users to identify without independent verification.

RealityCheck addresses this gap through a modular, six phase pipeline that operates externally and is model agnostic over existing LLMs. The system performs evidence grounded, claim level verification against Wikipedia sourced content and applies a multi signal verification framework combining NLI classification, semantic alignment, and rule based reasoning. Corrected responses are produced through deterministic answer synthesis without making any additional LLM calls or modifying the underlying model parameters. Each phase is independently replaceable and the pipeline supports multiple LLMs under identical experimental conditions.

Additionally, since RealityCheck operates without modifying the underlying model's parameters, it is a practically deployable system alongside existing state of the art LLMs, requiring no architectural rework from companies or developers. Additional evidence sources can further be integrated to broaden the scope of available information and make the system more viable at industry scale.

Experimental evaluation across IBM WatsonX Granite, Meta Llama, and MistralAI Mistral demonstrates significant improvements over raw LLM baselines across all reported metrics. Claim level accuracy, answer level accuracy, and hallucination recall improve consistently across all three models. The low overcorrection rate further confirms that the system does not aggressively rewrite LLM responses but intervenes selectively, only when backed by sufficient evidence.

Despite the rigorous experimentation and improvements demonstrated by the system, several limitations point toward productive directions for future work. The pipeline currently relies solely on Wikipedia as its knowledge source, limiting coverage of niche, domain specific, or temporally sensitive claims. Multi hop reasoning, where verification depends on chaining multiple intermediate facts, remains a challenge for the current atomic verification design. Retrieved evidence quality is also contingent on the quality of the claim's evidence query, which can degrade for short or heavily context dependent fragments. Future work should explore multi source evidence retrieval incorporating domain specific knowledge bases, stronger verification of compound reasoning over distributed evidence, adaptive evidence ranking, and domain specific grounding for high stakes applications.

Through this research, we conclude that hallucinations cannot be fully eliminated. But they can be significantly reduced when retrieval, verification, reasoning, and synthesis are treated as complementary stages within a unified pipeline. No single component, whether retrieval alone, NLI alone, or synthesis alone, can match the performance of the complete system as demonstrated by the ablation study. It is the compounded capability of each dedicated subsystem, designed to handle a distinct aspect of the verification problem, that enables the pipeline to operate effectively across diverse question types and model families.

As large language models continue to transition into high stakes real world deployments, externally grounded verification and correction frameworks such as RealityCheck may become increasingly important for enabling trustworthy human-AI interaction.

References

[1] Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.-t., Rocktäschel, T., Riedel, S., Kiela, D.: Retrieval-augmented generation for knowledge-intensive NLP tasks. In: Advances in Neural Information Processing Systems, vol. 33, pp. 9459–9474 (2020)

[2] Farquhar, S., Kossen, J., Kuhn, L., Gal, Y.: Detecting hallucinations in large language models using semantic entropy. Nature **630**, 625–630 (2024)

[3] Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y.J., Madotto,

A., Fung, P.: Survey of hallucination in natural language generation. ACM Computing Surveys **55**(12), 1–38 (2023)

[4] Dhuliawala, S., Komeili, M., Xu, J., Raileanu, R., Li, X., Celikyilmaz, A., Weston, J.: Chain-of-verification reduces hallucination in large language models. In: Findings of the Association for Computational Linguistics: ACL 2024, pp. 3563–3578. Association for Computational Linguistics (2024)

[5] Thorne, J., Vlachos, A., Christodoulopoulos, C., Mittal, A.: FEVER: a large-scale dataset for fact extraction and VERification. In: Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, pp. 809–819. Association for Computational Linguistics (2018)

[6] Guan, X., Liu, Y., He, H., Cheng, S., Li, W., Liang, J., Wen, J.-R.: Mitigating large language model hallucinations via autonomous knowledge graph-based retrofitting. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 38. AAAI Press (2024)

[7] Min, S., Krishna, K., Lyu, X., Lewis, M., Yih, W.-t., Koh, P.W., Iyyer, M., Zettlemoyer, L., Hajishirzi, H.: FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, pp. 12076–12100. Association for Computational Linguistics (2023)

[8] Semnani, S., Yao, V., Zhang, H., Lam, M.: WikiChat: Stopping the hallucination of large language model chatbots by few-shot grounding on Wikipedia. In: Findings of the Association for Computational Linguistics: EMNLP 2023, pp. 2387–2413. Association for Computational Linguistics (2023)

[9] Dey, P., Merugu, S., Kaveri, S.: Uncertainty-aware fusion: An ensemble framework for mitigating hallucinations in large language models. In: Companion Proceedings of the ACM Web Conference 2025. ACM (2025)

[10] Reimers, N., Gurevych, I.: Sentence-BERT: Sentence embeddings using siamese BERT-networks. In: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing, pp. 3982–3992. Association for Computational Linguistics (2019)

[11] Lin, S., Hilton, J., Evans, O.: TruthfulQA: Measuring how models mimic human falsehoods. In: Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pp. 3214–3252. Association for Computational Linguistics (2022)